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MTH213/MAT213: Differential Equation I

Solution of First Order ODEs: Method of Separation of Variables

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Solution of First Order ODEs

A first order ODE

$$\frac{dy}{dx} = F(x, y) \tag{1}$$

is called separable if $F(x, y)$ can be expressed as a product of a function of x and a function of y i.e.

$$\frac{dy}{dx} = g(x)h(y) \tag{2}$$

For example, the functions $e^{x+y} = e^x \cdot e^y$, $xy + 2x + 5y + 10 = (x + 5) \cdot (y + 2)$ are separable while $x^2 + 2xy + y^2$ is not.

Separable first order ODEs can be solved by the method of **separation of variables**.

Steps to Follow

Use the following steps to solve an ODE by the method of separation of variables:

Step 1: Separate the variables by moving terms involving variable y to the left-hand side and terms of variable x to the right-hand side:

$$f(y)dy = g(x)dx \tag{3}$$

Step 2: Integrating both sides of equation (3):

$$\int f(y)dy = \int g(x)dx, \tag{4}$$

Step 3: Add an arbitrary constant C on the right hand side.

Examples

1. Solve $\dfrac{dy}{dx} + \dfrac{1}{y}e^{-y^2+3x} = 0$

Solution

Separating variables yields: $-ye^{y^2}dy = e^{3x}dx$.

Integrating both sides: $-\int ye^{y^2}dy = \int e^{3x}dx$

$\therefore \quad -\dfrac{1}{2}e^{y^2} = \dfrac{1}{3}e^{3x} + C$ is the general solution.

Examples Contd.

2. Solve the IVP $y' = \dfrac{xy^3}{\sqrt{1+x^2}}, \quad y(0) = 1$

Solution

Separating variables yields: $y^{-3}dy = x(x^2+1)^{-\frac{1}{2}}dx$

Integrating both sides: $\int y^{-3}dy = \int x(x^2+1)^{-\frac{1}{2}}dx \therefore$

$-\dfrac{1}{2y^2} = \sqrt{1+x^2} + c$ is the general solution.

Applying the initial condition $-\dfrac{1}{2} = \sqrt{1} + c \quad \Rightarrow \quad c = -\dfrac{3}{2}$

The particular solution is $-\dfrac{1}{2y^2} = \sqrt{1+x^2} - \dfrac{3}{2}$.

Exercise

Solve the following ODEs

1. $(\sin x)(\sin y) + \cos y\, y' = 0$ 2. $xy^3 dx + (y+1)e^{-x} dy = 0$
3. $(3y^3 + 3y\cos y + 1)y' + \dfrac{(2x+1)y}{1+x^2} = 0$
4. $\tan y\, dx - \cot x\, dy = 0,\ \cos y \neq 0,\ \sin x \neq 0$
5. $\dfrac{dr}{d\theta} = \dfrac{r^2}{\theta},\quad r(1) = 2$
6. $\dfrac{dy}{dx} = \dfrac{2x(3\log x + 1}{\sin 2y + 3y\cos y}$